of-the-box—only the location where we place the code is different. Let's work with the default locations because configuring Apache is outside the scope of this article.

In Fedora, you write your code in */var/www/cgi-bin*, and in Ubuntu you write it in */usr/lib/cgi-bin*. The configuration files are in */etc/httpd* in Fedora and in */etc/apache2* in Ubuntu. Try the following code in *hello.py*:

```
#!/usr/bin/python
# Tell the browser that it is just plain text
# \n => a blank line. Indicates end of header
print 'Content-Type: text/plain\n'
# The content
print 'Hello, Friends!'
```

The first line instructs the shell to run this script using Python; but the file *hello.py* should be executable, that is:

```
chmod +x hello.py
```

Now, point your browser to *localhost/cgi-bin/hello.py* and you should see 'Hello, Friends!'.

Getting started with HTML

The Web pages are normally not plain text. They are HTML documents where the tags instruct the browser how to display the content. So, modify *hello.py* to be an HTML document.

```
#!/usr/bin/python
# Tell the browser that it is an html document
print 'Content-Type: text/html\n'
# Print the html document using triple quote for convenience.
print '''<html>
  <header></header>
  <body>
    <h3>Hello, Friends</h3>
    <p>Welcome - this is a paragraph.</p>
  </body>
</html>'''
```

The structure of the HTML document is <tag>content</tag>, where the content may include other tags. You are concerned with the body part. You now have two statements being displayed—one as a header and the other as a paragraph. You can try different types of headers—h1 through h4 and see the difference.

Displaying a form

You can now build a small application of classifieds for your friends. You will have three fields, a description and a contact e-mail ID. Write the following in *cgi-bin/form.py*:

```
#!/usr/bin/python
# Define the form
content_header = "Content-Type: text/html\n"
html_page = """
<html>
<head> </head>
<body>
<h1>Entry Form</h1>
<form method="post" action="form.py">
  <p><label>Title: </label><input type="text" name="title"
value="%(title)s"/ ></p >
  <p>Description: </p>
  <textarea name = "desc" rows=4 cols=60 >%(desc)s</textarea >
  <p><label>Email-id: <input type="text" name="email" value="%(email)s"
/></label></p >
  <button type="submit">Submit</button>
  <h4>%(message)s<h4>
</form>
</body></html>
"""
# Python Code
message = email = title = desc = ""
# present the form
print content_header
print html_page % {"title":title, "email":email, "desc":desc, "message":
message}
```

The string *html_page* is almost a static HTML page but with a few variables. The value for these variables is substituted in the print statement. Point your browser to *http://localhost/cgi-bin/form.py*.

You can enter the data, but not much happens. When you display the form initially, there will be no data in the form. The same program is being called when the form is 'actioned'. So, your code has to be able to differentiate between the two states. You will need to make use of the *cgi* module and add some Python code.

Adding logic

Add the following code after the Python code above:

```
import cgi
def process_form(email, title, desc):
  return "Your entry has been processed"
form = cgi.FieldStorage()
email = form.getvalue("email","")
title = form.getvalue("title","")
desc = form.getvalue("desc","")
if not email or not title:
  message = "Please enter Title and Email"
else:
  message = process_form(email, title, desc)
  email = title = desc = ""
# present the form
print content_header
print html_page % {"title":title, "email":email, "desc":desc, "message":
message}
```